

4 Dynamics of Expansion

4.1 Einstein's Field Equation

The dynamical equations of the evolution of the universe, *i.e.*, the Friedmann's equations follow from the Einstein's field equation

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi GT_{\mu\nu}. \quad (21)$$

where $R_{\mu\nu}$ is known as the curvature tensor, R is the curvature scalar, and $g^{\mu\nu}$ is the metric tensor. For our universe $g^{\mu\nu}$ is given by the FRW metric (discussed in the previous class).

Shortly after proposing the equation Einstein added a term Λ is called the *cosmological constant* to his field equation

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi GT_{\mu\nu}, \quad (22)$$

Einstein included the Λ term due to the following reason. Einstein proposed the field equation in 1915, which is almost 14 years before Hubble observed the expansion of the universe. Therefore, Einstein thought that the universe is static. But he noticed (based on inputs from the works of Alexander Friedmann, Georges Lemaître) that even in vacuum (*i.e.*, a universe where there is no mass and energy), the equation (21) predicts a universe that is dynamic. Therefore, he added the cosmological constant, such that for some value of Λ the field equation gives static solution.

After Hubble's discovery of the expansion of the universe Einstein regretted adding this term calling it his greatest blunder. But as we now know, he was correct. The universe has some cosmological constant. It is interpreted as the energy density of the vacuum. Such interpretation is motivated from the point of view of quantum theory which predicts that vacuum should have some energy. Quantum theory predicts that $\Lambda \sim m_{\text{Pl}}^4$, where $m_{\text{Pl}} = 10^{19}$ GeV is the Planck mass. But experimental observation of Λ is at least a factor of 10^{120} smaller. This is the largest known discrepancy in physics, and this problem is called the *cosmological constant problem*. Intuitively, the value of Λ is expected to be small since in the non-relativistic limit the field equation must give the Newton's 2nd law of gravitation.

4.2 Friedmann Equation

The evolution of the scale factor of the universe is given by the two Friedmann equations which follow from the Einstein's field equation (22)

$$H^2 = \frac{\dot{a}^2}{a^2} = \frac{8\pi G}{3}\rho + \frac{\Lambda}{3} - \frac{K}{a^2}, \quad (23)$$

and

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3P) + \frac{\Lambda}{3}. \quad (24)$$

The equations (23) and (24) are Friedmann equations in cosmology describing the expansion of the universe. The second equation, often called the acceleration equation tells us that the cosmological constant Λ accelerates the universe.

The two equations of Friedmann are not sufficient to determine the dynamics alone and one also needs to know how the mass and energy density evolves. The evolution of the energy density is given by the fluid equation

$$\frac{\partial \rho}{\partial t} + \frac{\dot{a}}{a} \left(3\rho + 3P \right) = 0. \quad (25)$$

By defining

$$\omega = \frac{P}{\rho}.$$

it can be shown that the fluid equation as a solution of the type

$$\boxed{\rho \propto a^{-3(1+\omega)}}. \quad (26)$$

The equation is known as the equation of state.

The three types of matter frequently encountered in the universe are

- **Cold matter/dust/non-relativistic matter:** By definition, the momentum of a non-relativistic particle is negligibly small, so that for non-relativistic matter the pressure $P = 0$, and $\omega = 0$, so that the equation of state takes the form

$$\rho \propto a^{-3}. \quad (27)$$

- **Radiation/hot matter:** Radiation is characterized by masslessness. In cosmology, radiation includes photons, neutrinos etc. The pressure for radiation is given by $P = \rho/3$, so that the equation of state takes the form

$$\rho \propto a^{-4}. \quad (28)$$

- **Cosmological constant:** Taking the cosmological constant term to the right hand side, the field equation reads

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi GT_{\mu\nu} - \Lambda g_{\mu\nu}.$$

So, the $\Lambda g_{\mu\nu}$ term also acts as a source of gravity. Therefore, it makes sense to consider it a different type of matter/energy – can be interpreted as the energy associated with the vacuum. In fact, this type of matter/energy is unusual, because (following the previous derivation) (*given as exercise*)

$$-P = \rho_\Lambda = \frac{\Lambda}{8\pi G}. \quad (29)$$

This is a matter/energy with negative pressure. This is exactly what Einstein wanted to have a matter that counters the effect of gravity. So the equation of state for this type of matter is $\omega = -1$.

$$\rho_\Lambda = \text{constant}.$$

Absorbing the ρ_Λ into ρ the Friedmann equations can be written as

$$\boxed{H^2 = \frac{\dot{a}^2}{a^2} = \frac{8\pi G}{3}\rho - \frac{K}{a^2}}, \quad (30)$$

and

$$\boxed{\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3P)}. \quad (31)$$

4.3 Critical density and K

Rearranging (30) we get

$$1 = \frac{\rho}{3H^2/8\pi G} - \frac{K}{a^2H^2} = \frac{\rho}{\rho_{\text{crit}}} - \frac{K}{a^2H^2}.$$

This equation tells that for any value of H , *i.e.*, at any epoch of the cosmic evolution, there is a critical density

$$\rho_{\text{crit}} = \frac{3H^2}{8\pi G}, \quad (32)$$

of the universe such that if the energy density of the universe ρ is

- $\rho = \rho_{\text{crit}}$ then $K = 0$, the universe is flat

- $\rho > \rho_{\text{crit}}$ then $K > 0$, the universe is closed
- $\rho < \rho_{\text{crit}}$ then $K < 0$, the universe is open

At the present epoch t_0 , the Hubble constant is $H_0 = 72 \text{ km/s/Mpc}$ and the corresponding critical density is

$$\rho_{\text{crit},0} = \frac{3H_0^2}{8\pi G} = 1.9h^2 \times 10^{-29} \text{ g cm}^{-3}, \quad (33)$$

where the number to the left is obtained by substituting the present Hubble parameter $H_0 = 100h \text{ km/s/Mpc}$ where $h = 0.72$.

There is something strange associated with the critical density of the universe. The energy density ρ of mass and radiation scales as $1/a^3$ and $1/a^4$, respectively, while the energy density associated with Λ remains constant. So, if we extrapolate the Friedmann equation (23) to early times, then ρ would be large and dominated by radiation. One can then neglect the $-K/a^2$ term which is negligible compared to $1/a^4$ or $1/a^3$ terms. Hence

$$H^2 = \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho \Rightarrow \boxed{1 = \frac{\rho}{\rho_{\text{crit},0}}}. \quad (34)$$

It means that the density of the universe was equal to its critical density at very early times. Experimental observations show that even today the density of the universe ρ_0 is fairly close to the critical density $\rho_{0,\text{crit}}$ implying a flat $K = 0$ universe. This is a puzzle because one normally expects that billion years of evolution would make ρ_0 different from $\rho_{\text{crit},0}$. It referred to as the *flatness problem*.

4.4 Model Solutions

In this section we discuss a few model solutions of Friedmann equations where the solutions are exact.

4.4.1 Einstein Static Universe

A static universe implies that $\dot{a} = 0$ and $\ddot{a} = 0$. From the Friedmann equation (30) we can write

$$0 = \frac{8\pi G}{3}\rho - \frac{K}{a^2}, \Rightarrow \frac{8\pi G}{3}\rho = \frac{K}{a^2}.$$

As ρ is positive K must be equal to $+1$. The acceleration equation gives

$$\rho + 3P = 0,$$

which implies the pressure must be negative. So we can consider a new kind of entity with some associated energy density. We see that the cosmological constant fit the bill. It has energy density ρ_Λ and pressure $P = -\rho_\Lambda$. The fluid equation can be re-written as

$$\rho_m + \rho_\Lambda - 3\rho_\Lambda = 0, \quad \rho_m = 2\rho_\Lambda.$$

Hence the scale factor is

$$a = \frac{1}{\sqrt{4\pi G\rho_m}} = \frac{1}{\sqrt{\Lambda}}.$$

4.4.2 Matter Dominated Universe

If the universe is matter dominated, then $\Lambda = 0$, and the energy density $\rho \propto 1/a^3$. Denoting the density at the present time t_0 by $\rho_0 \equiv \rho(t_0)$, we can write

$$\rho = \rho_0 \left(\frac{a_0}{a}\right)^3,$$

where a_0 is the scale factor at the present time t_0 $a_0 = a(t_0)$. The first Freidmann equation (30) as

$$\begin{aligned}\dot{a}^2 &= \frac{8\pi G}{3}\rho_0\left(\frac{a_0}{a}\right)^3 a^2 - K, \\ &= \frac{2A}{a} - K, \quad A = \frac{4\pi G}{3}\rho_0 a_0^3\end{aligned}\quad (35)$$

To solve it, it is convenient to introduce a new variable η , called the *conformal time*

$$\frac{d\eta}{dt} = \frac{1}{a}. \quad (36)$$

This definition of the conformal time does not fix the constant of the integration. Unless we are studying physics of inflation, $\eta = 0$ at the time of the big bang $t = 0$. The conformal time is not the age of the universe. If the conformal time since the big bang is multiplied by the speed of light, it gives the total comoving distance that is traveled by a light ray. In terms of conformal time, the equation (35) becomes

$$\left(\frac{da}{d\eta}\right)^2 = 2Aa - Ka^2.$$

This equation can be integrated from the time of the big bang $a = 0$ to any time

$$\eta = \int_0^a \frac{da}{\sqrt{2Aa - Ka^2}}. \quad (37)$$

Lets consider solutions for three different scenarios

For $K = 0$: In this case the solution is

$$a = \frac{4\pi G}{3}\eta^2,$$

and

$$t = \int a d\eta = \int_0^\eta \frac{4\pi G}{3}\eta^2 d\eta = \frac{4\pi G}{9}\eta^3.$$

These two parametric solutions can be combined to give

$$a = \left(\frac{4\pi G}{3}\right)^{1/3} (3t)^{2/3}. \quad (38)$$

The matter dominated ($\Lambda = 0$) flat universe is called the Einstein-de Sitter universe.

For $K = -1$ the integration eq.(37) is

$$\eta = \int_0^a \frac{da}{\sqrt{2Aa + a^2}}, \quad a = \frac{8\pi G}{3}(\cosh \eta - 1), \quad t = \frac{8\pi G}{3}(\sinh \eta - \eta). \quad (39)$$

For $K = 1$ the integration eq.(37) is

$$\eta = \int_0^a \frac{da}{\sqrt{2Aa - a^2}}, \quad a = \frac{8\pi G}{3}(1 - \cos \eta), \quad t = \frac{8\pi G}{3}(\eta - \sin \eta). \quad (40)$$

The variation of scale factor with cosmological time is shown in figure 14. As can be seen from the figure, for $K = 1$ the universe expands to a mximum size and then collapses to $a = 0$. In the $K = -1$ open universe, the universe expands for ever, where as $K = 0$ is an intermediate scenario between the two.